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ESIS: Edge Survival Intelligence System

Offline Decision Intelligence for Human Survival Under Constraints

Motivation: The Crisis of Survival Under Constraint

In moments of acute crisis—housing instability, disaster response, or economic collapse—individuals are forced to make high-stakes decisions with limited information, depleted resources, and often no internet access. Paradoxically, these are the exact conditions under which modern AI systems fail. Today's flagship AI models are cloud-dependent, require persistent connectivity, and are designed primarily as general-purpose chatbots. This creates three critical limitations:

- **Connectivity Dependence:** AI is least available when it is most needed. In disaster zones, rural environments, or periods of instability, internet access is unreliable or nonexistent.
- **Lack of Decision Intelligence:** Existing AI tools provide generalized advice but lack the ability to evaluate constrained trade-offs involving money, time, and risk.
- **Privacy and Trust Constraints:** Vulnerable populations cannot safely transmit sensitive data—financial, medical, or legal—across cloud systems. Privacy is not optional in these contexts; it is foundational.

This creates a fundamental gap: **the people who most need intelligent systems are the least able to access them**.

ESIS (Edge Survival Intelligence System) is designed to close this gap by delivering offline, private, and decision-grade intelligence directly to the user. It is not just a technical project, it is a mission to ensure that no one is left alone in the cold, wet, and dangerous streets, deprived of the resources and support that should be a human right.

Core Concept: Decision Optimization Under Uncertainty

ESIS is a **local-first AI system powered by Gemma 4** that enables users to make optimal decisions under real-world constraints. Unlike traditional AI assistants, ESIS does not operate as a conversational interface. It functions as a **decision optimization engine under uncertainty**. The system is designed to:

- Allocate scarce resources (money, time)
- Prioritize actions under risk and urgency
- Evaluate trade-offs between competing needs
- Identify critical support pathways
- Quantify outcomes before decisions are made

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ESIS transforms AI from passive guidance into **active, structured decision intelligence**. It is engineered for the hardest moments—when every dollar, every hour, and every choice can mean the difference between survival and catastrophe.

Technical Architecture: Integrating Local AI, Decision Modeling, and Simulation

1. Local Reasoning via Gemma 4

At the core of ESIS is a **locally deployed Gemma 4 model**. This enables:

- **Offline inference** without external dependencies
- **Low-latency responses** suitable for real-time decision-making
- **Privacy-preserving computation**, ensuring sensitive data never leaves the device

Gemma 4 is not used as a generic chatbot, but as a **structured reasoning engine** that produces actionable, constrained outputs. Its support for **native function calling** and **multimodal input** (text, image, audio) makes it uniquely suited for this application.

2. State-Based Decision Modeling

ESIS represents the user's situation as a **structured state vector**, for example:

python
```
state = {
    "cash": 40.0,
    "days_to_instability": 3,
    "food_security": 0.3,
    "housing_risk": 0.8,
    "health_risk": 0.2,
    "network_support": 0.1
}
```

This abstraction allows the system to reason explicitly about:

- Resource depletion over time
- Risk escalation
- Trade-offs between competing priorities

By converting real-world conditions into a formal state space, ESIS enables both deterministic planning and probabilistic evaluation.

3. Action Space and Optimization

Each possible user decision is treated as an **action**:

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- Spend vs conserve resources
- Seek assistance vs delay
- Relocate vs remain
- Prioritize food vs housing vs health

ESIS evaluates these actions using an **expected utility framework**:

$$a^* = \arg\max_a \mathbb{E}[R(s,a)]$$

Where:

- s is the current state
- a is a candidate action
- R is a survival-oriented reward function

This transforms decision-making from reactive advice into **quantitative optimization under constraints**.

4. Monte Carlo Simulation Layer

To capture uncertainty, ESIS performs **forward simulations** over possible outcomes. For each candidate action, the system estimates distributions of outcomes:

- Probability of stability
- Probability of risk escalation
- Probability of failure (e.g., loss of housing, resource exhaustion)

This enables outputs such as:

- "Spending $40 today results in a 68% chance of short-term stability but increases long-term risk."
- "Delaying action improves survival probability by 22% over a 3-day horizon."

This simulation layer is critical—it allows users to understand consequences before committing to decisions.

5. Function Calling and Action Integration

Gemma 4's **native function calling capability** is used to connect reasoning with real-world actions. The model can trigger structured operations such as:

- Resource lookup (e.g., nearby shelters, food banks, clinics)
- Plan generation (prioritized next steps)
- Risk evaluation (simulation outputs)

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This transforms the system from passive advisor to **active decision assistant**.

6. Offline Data Integration

ESIS includes **preloaded datasets** relevant to survival scenarios:

- Food assistance locations
- Shelter availability
- Medical clinics
- Legal and social services

Because these datasets are stored locally, the system remains functional even without internet access. Periodic updates can be synchronized when connectivity is available.

7. Multimodal Capability (Optional Extension)

Leveraging Gemma 4's multimodal capabilities, ESIS can extend beyond text input:

- **Image understanding** (e.g., identifying documents or notices)
- **Voice interaction** for accessibility
- **Contextual interpretation** of real-world inputs

This further enhances usability in constrained environments.

Mathematical Framing: Markov Decision Process and Monte Carlo Simulation

ESIS is not just a heuristic tool—it is grounded in **applied stochastic control** and **Markov Decision Process (MDP)** theory. The core objective is:

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right]$$

Where:

- s_t= survival state at time t
- a_t= action (spend, move, call, apply)
- R= survival utility (housing stability, food, safety)
- γ= discount factor (captures urgency/priority)

Constraints:

- $\text{cash}_t \geq 0$

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- $\text{risk}_t \leq \tau$(risk threshold)

Interpretation: This is essentially a **CondorBrain™ for human survival**—a system that, through millions of Markov-Monte Carlo iterations, finds the path of least resistance and immediately guides the person to safety, food, and critical resources.

Repository Outline and Implementation Strategy

Directory Structure

plaintext

```
esis/
├── core/
│   ├── gemma_inference.py         # Gemma 4 local inference engine
│   ├── state_model.py             # State vector and transition logic
│   ├── decision_engine.py         # Action evaluation and optimization
│   ├── simulation.py              # Monte Carlo simulation engine
│   ├── function_calls.py          # Structured tool integration
│   └── multimodal.py              # Image/voice input processing
├── data/
│   ├── shelters.json              # Preloaded shelter data
│   ├── food_banks.json            # Preloaded food bank data
│   ├── clinics.json               # Preloaded medical clinic data
│   ├── legal_aid.json             # Preloaded legal aid data
│   └── offline_maps/              # Vector tiles for offline routing
├── ui/
│   ├── cli.py                     # Command-line interface
│   └── simple_ui.py               # Minimal GUI (optional)
├── tests/
│   ├── test_state_model.py
│   ├── test_decision_engine.py
│   ├── test_simulation.py
│   └── test_function_calls.py
├── scripts/
│   ├── setup_env.sh               # Environment setup script
│   ├── run_esis.py                # Main entry point
│   └── update_data.py             # Offline data pack updater
├── docs/
│   ├── architecture.md
│   ├── usage.md
│   └── hackathon_writeup.md
├── requirements.txt
└── README.md
```

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Key Scripts and Their Functions

Script/Module	Functionality
`gemma_inference.py`	Loads and runs Gemma 4 locally (via Ollama, llama.cpp, or vLLM). Handles text and multimodal input.
`state_model.py`	Defines the state vector, state transitions, and encodes user context.
`decision_engine.py`	Implements action space, reward function, and optimization logic (MDP solver).
`simulation.py`	Runs Monte Carlo simulations to estimate outcome probabilities for candidate actions.
`function_calls.py`	Integrates Gemma 4 function calling with local tools (resource lookup, plan generation, etc.).
`multimodal.py`	Handles image (OCR), document, and voice input for accessibility and document understanding.
`cli.py`	Provides a command-line interface for user interaction.
`setup_env.sh`	Sets up the Python environment and installs dependencies.
`update_data.py`	Updates offline data packs (shelters, food banks, clinics) from authoritative sources.

Intensive Math, Derivations, and Terms Defined

Markov Decision Process (MDP) Formulation

- **State (s)**: Encodes the user's current survival context (cash, food, housing risk, health, etc.).
- **Action (a)**: A discrete or continuous choice (e.g., spend $X, seek shelter, delay action).
- **Transition ($P(s' \mid s, a)$)**: Probability of moving to state s'after taking action $a$in state s.
- **Reward ($R(s, a)$)**: Quantifies the utility of survival, stability, or risk reduction.
- **Policy (π)**: Mapping from states to actions.
- **Objective**: Maximize expected cumulative reward over a planning horizon T:

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{T} \gamma^t R(s_t, a_t)\right]$$

- **Constraints**: Enforce resource limits and risk thresholds.

Monte Carlo Simulation

For each candidate action a:

1. **Sample** Ntrajectories by simulating state transitions under stochastic dynamics.
2. **Estimate** outcome probabilities (e.g., stability, risk, failure) empirically.
3. **Aggregate** results to inform the user:

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$$\text{Outcome Distribution} = \begin{cases} \text{Stable:} & 68\% \\ \text{At Risk:} & 22\% \\ \text{Failure:} & 10\% \end{cases}$$

Function Calling and Tool Integration

Gemma 4's function calling is mapped to local Python functions, for example:

```
def find_nearest_shelter(location):
    # Returns list of shelters sorted by distance and availability
    pass

def generate_survival_plan(state):
    # Returns prioritized actions for next 24-72 hours
    pass

def run_risk_simulation(state, action):
    # Returns outcome probabilities for given action
    pass
```

Multimodal Processing

- **OCR**: Extracts text from images of documents (e.g., eviction notices) using on-device OCR (e.g., PaddleOCR, Tesseract, or LLM-based OCR).
- **ASR**: Converts speech to text for voice-first interaction.
- **Image Understanding**: Classifies and interprets document types for context-aware guidance.

Data Integration: Offline Data Packs and Authoritative Sources

ESIS relies on **offline data packs** for critical resources:

- **Shelters**: Locations, capacity, eligibility, hours
- **Food Banks**: Locations, schedules, requirements
- **Medical Clinics**: Walk-in hours, urgent care, triage protocols
- **Legal Aid**: Contact info, intake procedures
- **Maps**: Offline vector tiles for routing and navigation

Data Sourcing and Curation:

- Data is sourced from authoritative public datasets (e.g., HUD, FEMA, local 211 directories, government open data portals).
- Data packs are curated, validated, and periodically updated via the `update_data.py` script.
- All data is stored locally to ensure full offline functionality.

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Multimodal Interaction: Accessibility and Real-World Usability

ESIS is designed for **maximum accessibility**:

- **Voice-First UI**: Enables users with low literacy or visual impairment to interact via speech.
- **Image Input**: Users can photograph documents (e.g., eviction notices), and the system will extract and explain the content.
- **Low-Resource UI**: Minimalist interface for low-end devices, with battery and data usage optimization.

Privacy-Preserving Methods and Ethics

ESIS is architected for **privacy by design**:

- **No cloud dependency**: All inference and data processing occurs locally.
- **No telemetry or data exfiltration**: User data never leaves the device.
- **Differential privacy and federated learning** (future extension): If community-level learning is desired, only aggregated, noise-added updates are shared, never raw data.
- **Ethical guardrails**: The system is transparent about its limitations and never makes irreversible or high-risk recommendations without explicit user confirmation.

Medical Triage and Clinical Decision Support

ESIS can be extended to provide **offline medical triage**:

- **Symptom input**: Users describe symptoms via text or voice.
- **Rule-based triage**: The system applies offline clinical rule sets to classify risk (e.g., urgent, non-urgent, self-care).
- **Safety constraints**: Always errs on the side of caution; when in doubt, recommends seeking in-person care.

Geolocation, Offline Maps, and Routing

- **Offline maps**: Vector tiles and routing data are preloaded for the target region.
- **Geolocation**: Uses device GPS (if available) to locate nearest resources.
- **Routing**: Provides step-by-step directions to shelters, food banks, or clinics without internet.

Evaluation Metrics and Simulation-Based Validation

ESIS is validated using **simulation-based testing**:

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- **Scenario coverage**: Tests include a range of crisis scenarios (e.g., sudden loss of housing, medical emergency, food insecurity).
- **Outcome metrics**: Measures include survival probability, risk reduction, and resource utilization.
- **Safety testing**: All recommendations are checked for safety and feasibility before presentation.

Hardware Targets and Performance Optimization

ESIS is engineered for **edge deployment**:

- **Supported hardware**: Laptops, desktops, Jetson Nano/Orin, Raspberry Pi, and portable SSDs.
- **Model selection**: Automatically chooses the optimal Gemma 4 variant (E2B, E4B, 7B, 14B) based on available RAM and compute.
- **Quantization**: Uses 4-bit, int8, or NVFP4 quantized checkpoints for memory efficiency.
- **Inference runtimes**: Supports Ollama, llama.cpp, vLLM, and Unsloth for efficient local inference.

Model Compression, Distillation, and Domain Adaptation

- **Compression**: Applies quantization and pruning to fit models on low-resource devices.
- **Distillation**: Fine-tunes Gemma 4 on survival-specific data for improved relevance.
- **LoRA**: Uses Low-Rank Adaptation for efficient domain adaptation without full retraining.

Legal, Liability, and Regulatory Considerations

- **Disclaimer**: ESIS is not a substitute for professional medical, legal, or emergency services.
- **Liability**: Clear user agreements and disclaimers are provided.
- **Regulatory compliance**: Data sources and processing comply with relevant privacy laws (e.g., GDPR, HIPAA).

Partnerships, Deployment Pathways, and Community Engagement

- **NGOs and Shelters**: Collaborate to distribute ESIS on portable SSDs or preloaded devices.
- **Local Governments**: Partner for data updates and outreach.
- **Community Champions**: Train local volunteers to assist with deployment and support.

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Storytelling, Video Production, and Hackathon Submission Strategy

Video Script Outline

1. **Hook**: "What would I do if I had $40, no internet, and 3 days before losing everything?"
2. **Problem**: Show the reality of crisis—no access to information, bad decisions, and the risk of collapse.
3. **Solution**: Demonstrate ESIS running offline—inputting a scenario, generating a plan.
4. **WOW Moment**: Show simulation—"If you spend today → 10% chance of failure. If you wait → 65% stability."
5. **Impact**: "This system works anywhere. No internet. No cloud. Just advanced AI reasoning saving lives."

Submission Writeup

- **Motivation**: Frame the problem emotionally and technically.
- **Constraint**: Emphasize offline limitation and privacy.
- **Design Philosophy**: "We do one thing well—decision intelligence under constraint."
- **Technical Depth**: Detail the architecture, math, and simulation.
- **Impact**: Highlight real-world applicability and transformative potential.

Competitive Differentiators

- **Not a chatbot**: ESIS is a decision engine, not a conversational assistant.
- **Offline-first**: Operates fully without internet—critical for disaster and survival scenarios.
- **Quantitative reasoning**: Uses MDPs and Monte Carlo simulation for risk-aware planning.
- **Privacy-preserving**: No data leaves the device.
- **Multimodal**: Supports text, image, and voice input for accessibility.
- **Real-world datasets**: Integrates authoritative, curated offline data packs.

Potential Extensions and Future Directions

- **Federated learning**: Enable community-level learning without compromising privacy.
- **Medical triage**: Expand clinical decision support with offline rule sets.
- **Community mesh**: Devices can sync data and updates opportunistically when in proximity.
- **Global deployment**: Localize data packs and language support for international use.

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Impact Narrative: Why ESIS Matters

ESIS is not just a technical achievement, it is a **lifeline** for those who have been left behind by the digital revolution. It is for the person sitting alone on a freezing street, for the family facing eviction, for the survivor of disaster with no access to help. It is a system that says: **"You are not alone. You have options. You have advanced reasoning, planning and intelligence at your side."** This project is a call to action for the AI community, the press, and society at large. It is a demonstration that **AI can be a force for equity, resilience, and survival—not just convenience or profit**. It is a blueprint for how we can reclaim credibility and leadership in AI for social good, by building systems that matter when everything else fails.

Conclusion

ESIS redefines what is possible for AI at the edge. By combining Gemma 4's local reasoning capabilities with structured decision modeling and probabilistic simulation, it transforms AI into a system capable of supporting real-world survival decisions—offline, private, and actionable. This approach is not limited to a single domain. It establishes a general framework for deploying decision intelligence in resource-constrained environments, with applications spanning humanitarian aid, public health, disaster response, and beyond.

ESIS demonstrates that when advanced models are made local, private, and actionable, AI becomes not just accessible—but essential.

Final Note

This project is not just for the competition; the passion behind it is not for the prize money, its not for street-cred, and not for bragging rights. Been there, done that, and it got me nothing but hardship, emotional and physical pain, false imprisonment, and dropping dead in the streets on November 16, 2025 where I was fully resuscitated, and spent 7 weeks in the hospital recovering, and discharged on December 31, 2025, back to homelessness, but not to the streets this time. I have been working on novel approaches to advanced neural nets, and one that shows promise for the first of its kind, the ability to self learn without programming and self detect and correct malfunctions, and develop cognitive skills with a simple set of innate characteristics. Much like a baby has innate characteristics, so shall CondorNet v6.0 (about 12 to 24 months out). Parts of CN v4.6 and CondorBrain™ abd CondorIntelligence™ first specifically designed and developed for FinTech Automated Options Trading, parts of its core are being used in ESIS running on Gemma 4.

Now that we know what its primary purpose does not serve, of more importance, ESIS is for those who have been left with nothing, and have nothing left in life but to sit on a street corner and drop dead, with but a few breaths left, and a femtometer left in their spirit, of hope for survival. ESIS is a testament to what AI can and should be: **a tool for dignity, resilience, and life itself**. ESIS is for GOD's children, I being a child of our living God, YHWH, and as well as

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a Physicist and Software Engineer and former Chairman and CEO of software tech companies for > 27 years in my previous life, I am also an ordained Christian Pastor.

In Acts 14:22 – "We must all endure many hardships before entering the Kingdom of GOD…" The key word there is endure.

We cannot do so with the current homeless systems in place. We can with ESIS, and with faith.

GOD is not happy with how his Children are being treated and how we are suffering. He also commands that we clothe, feed and provide shelter to the Homeless, the disabled, the injured, those in a weakened state, because it is in such a weakened state, that we are the strongest, as that is when the Holy Spirit is strongest - in and around us and standing for those who stand strong against the forces of darkness. Let's build a future where no one is left behind, and one where GOD's grace alone is beyond sufficient for us to make GOD #1 in our lives. Open your hearts, and pray, and you will see the vision gifted me by the Holy Spirit directly into my heart.

I am here to let you know GOD is very closely watching, we are all always under the tests of the wicked, this test, is a test of how you decide to resolve a critically serious global-scale set of issues, some of which may very well be an abomination to GOD, especially when the Holy Spirit is right here, right now*, awaiting your decision to help accelerate this on a global scale*. You really have two choices here: Jump back on GOD's pathway, OR don't do so. No one foot in one foot out of the doorway, that shows lack of faith. Be Peter when he focused on Christ and stepped out of the boat and walked on water for a second until he took his eyes off of Christ and immediately submerged. If you knew GOD, as in Jehovah, was paying close attention to you or Yeshua (Christ), was standing behind you, how would you decide on this by comparison to the competitors? I got a secret for you, Christ is and always will be right behind you, for the Kingdom of Heaven is upon us. Do we make the same mistakes again as history does repeat itself. Or do we love one another, even our enemies, for there is a special place in Heaven for those who can love their enemies. *I, probably won't make it to that special place.*

I am here to complete this mission and do so from love in my heart for those in need. That is why GOD allowed to have happen what happened to me. That is why I am right here right now, as I now fully have seen, felt and understand the miserable life of those in such a terrifying and horrible position, and to know and feel how they are mistreated.

Ask yourselves one question, "If I was taken away, falsely imprisoned (every apostle of Christ, John the Baptist, and Christ himself were imprisoned falsely), and everything lost, stolen, and my close friends and family rejecting me much like Bildad, Eliphaz and Zophar did to Job, and the intensive betrayal I had to endure my heart being staked a thousand times over and over again for many years, yet, my faith is stronger than ever before…How then, would you conclude that GOD does not have his hand in on this? To do so, would be illogical, and not on GOD's pathway. And if we are not on GOD's pathway, whose pathway are we on? Rhetorical question.

May the Grace of GOD be with you and your families. Amen.

/Pastor T. Jerry Mahabub

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Appendix A: Example Use Case Scenario

Scenario: A user has $40, three days before losing housing, and no internet access.

ESIS Workflow:

1. **Input:**
   - Cash: $40
   - Days to instability: 3
   - Food security: low
   - Housing risk: high
2. **State Vector Construction:**
   - `state = {"cash": 40, "days_to_instability": 3, "food_security": 0.3, "housing_risk": 0.8}`
3. **Action Generation:**
   - Spend $20 today
   - Seek shelter immediately
   - Delay spending, conserve cash
4. **Monte Carlo Simulation:**
   - Spend $20 today: 62% stable, 28% high risk, 10% failure
   - Delay spending: 68% stable, 22% risk, 10% failure
5. **Recommendation:**
   - "Reduce spend today. Seek shelter at [nearest location]. Maximize cash buffer. Contact [outreach]."
6. **Function Calls:**
   - `find_nearest_shelter(location)`
   - `generate_survival_plan(state)`
   - `run_risk_simulation(state, action)`
7. **Output:**
   - Prioritized action plan for next 48 hours
   - Directions to nearest shelter
   - Outreach message template

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Appendix B: Repository Quickstart

1. **Clone the repository:** `git clone https://github.com/your-org/esis.git`
2. **Set up environment:** `bash scripts/setup_env.sh`
3. **Run ESIS:** `python scripts/run_esis.py`
4. **Update data packs:** `python scripts/update_data.py`
5. **Run tests:** `pytest tests/`